

**TRƯỜNG ĐẠI HỌC CÔNG NGHIỆP TP.HCM**

**Khoa Công Nghệ Điện Tử - Bộ môn Điện Tử Máy Tính**

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**BÀI GIẢNG**

**HỆ THỐNG NHÚNG  
(Embedded System)**

**TP.HCM-2024**

# NỘI DUNG MÔN HỌC

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**Chương 1: GIỚI THIỆU HỆ THỐNG NHÚNG**

Chương 2: KỸ THUẬT GIAO TIẾP NGOẠI VI

Chương 3: CẤU TRÚC PHẦN MỀM

Chương 4: THIẾT KẾ HỆ THỐNG NHÚNG

# CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

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1. HỆ THỐNG NHÚNG
2. PHẦN CỨNG HỆ THỐNG NHÚNG
3. PHẦN MỀM HỆ THỐNG NHÚNG
4. CÔNG NGHỆ Multi-Core, SoC, and FPGA
5. MỘT SỐ ỨNG DỤNG HỆ THỐNG NHÚNG

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

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- ☐ Multiprocessors System
- ☐ Multi-core Processor
- ☐ Graphic Processing Units (GPUs)
- ☐ Multiprocessor System on Chip (MPSoC)
- ☐ Field Programmable Gate Arrays (FPGAs)

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.1 Multiprocessor System

### 4.1.1 The Needs of Multiprocessors

- A **multiprocessor** is, in general, any computer system with two or more processors coupled together.
- The reasons for multiprocessors are the same reasons that drive all of embedded system design:
  - Real-time performance
  - Power consumption
  - Cost

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.1 Multiprocessor System

- The **cost** of a microprocessor increases greatly when *the clock speed increases*. Because the *fastest processors are very costly*, **splitting** the application so that it can be performed on **several smaller processors** is much *cheaper*.
- In addition to reducing costs, using multiple processors can also help with **real-time performance**. → *meet deadlines* and *be responsive to interaction* much more easily.
- Several processors running at slower clock rates **consume less power** than *a single large processor*.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.1 Multiprocessor System

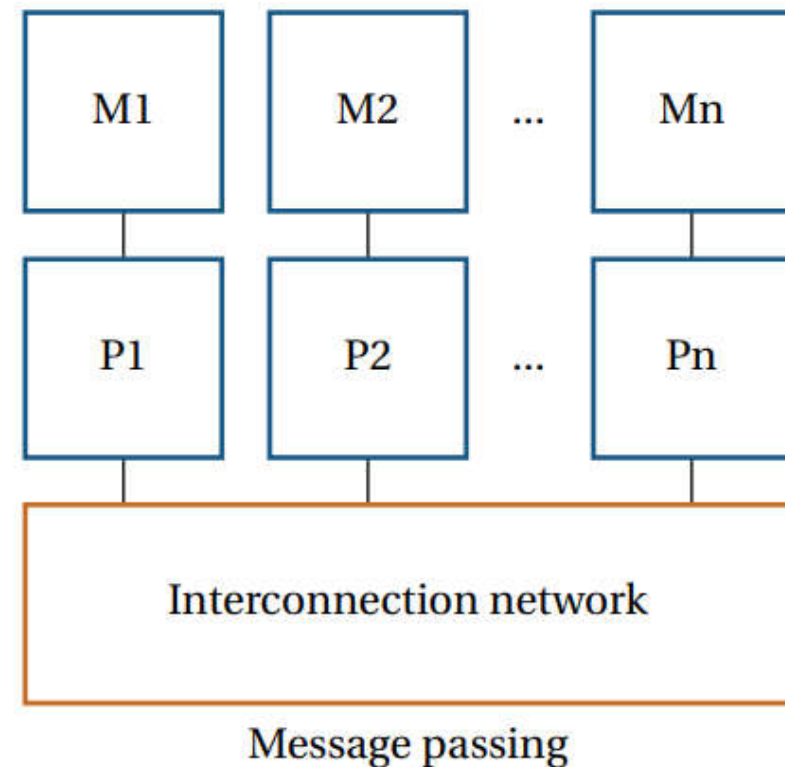
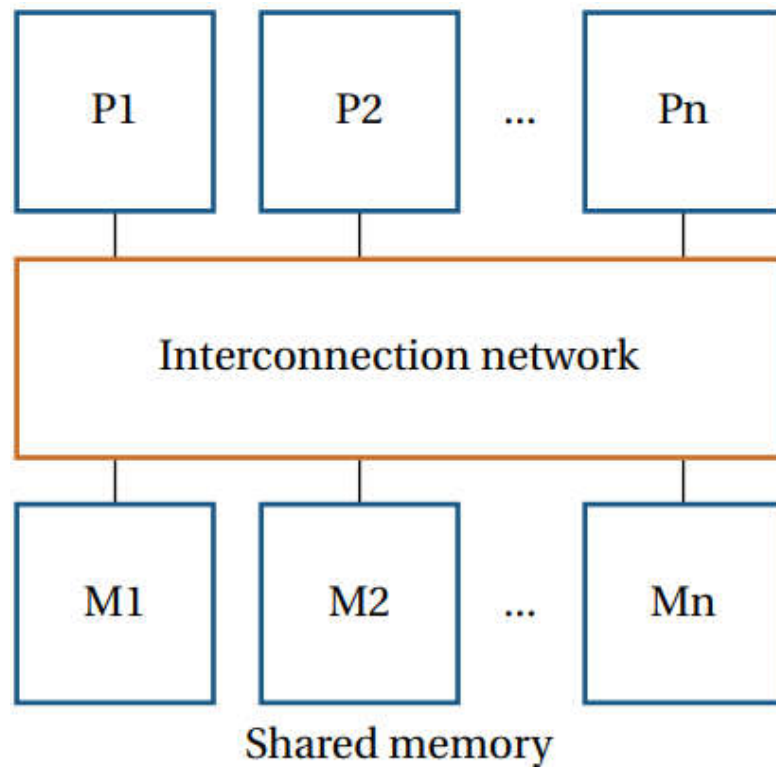
### 4.1.2 Categories of Multiprocessors

- + The **multiprocessing** has been used for relatively low-performance systems and to achieve **very high levels of real-time performance** at **very low energy levels**.
- + There are two major types of multiprocessor architectures:
  - **Shared memory systems** have a pool of processors that can read and write **a collection of memories**.
  - **Message passing systems** have a pool of processors that can **send messages** to each other. Each processor has its **own local memory**.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.1 Multiprocessor System

### 4.1.2 Categories of Multiprocessors



# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.1 Multiprocessor System

### 4.1.2 Categories of Multiprocessors

- **Shared memory systems:** are very common in **single-chip embedded multiprocessor systems** → offers *relatively fast access* to shared memory.  
→ It shows up in **low-cost** systems (*CD players*) and also appear in **higher-cost, high-performance** systems (*cell phones*).
- **Message passing systems:** are widely used in **distributed embedded systems** (*Example of a safety-critical real-time distributed embedded system is found in the automobile*).

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.1 Multiprocessor System

### ➤ Distributed system

+ The term distributed system is used for a **multiprocessor**, in which the *processing elements are physically separated* → These systems can be modeled as **message-passing systems** (CAN bus, Internet Protocol,..)

### ➤ Multiprocessor system-on-chip (MPSoC)

+ In contrast, a multiprocessor system-on-chip is a system-on-chip (**SoC**) with *multiple processing elements*.

+ Shared memory processors are well-suited to applications that require a **large amount of data** to be processed. **Signal processing systems stream data** and can be well-suited to shared memory processing.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.2 Multi-core Processors

- The move to **multicore processor** architectures in the embedded processor industry has been spurred by other trends occurring over the past few years as well, including:
  - The micro-architectural techniques employed to extract *parallelism* from one instruction stream are *becoming more expensive* relative to the *performance gain*.
  - Clock scaling *reaching limits* due to *power constraints* – an empirical study suggests that *a 1% clock speed increase* results in *a 3% power increase*.

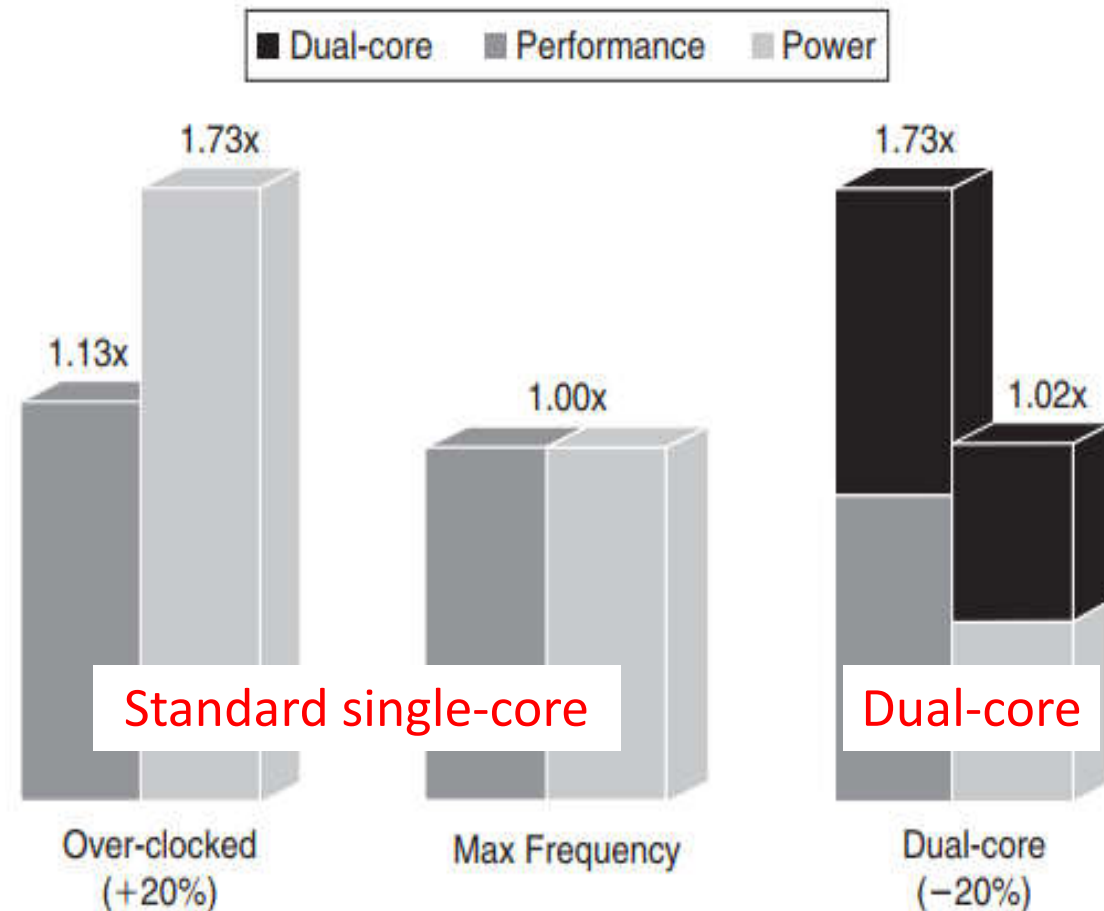
# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.2 Multi-core Processors

### 4.2.1 Motivation

+ **Figure** provides further insight into why **multi-core processing** is particularly attractive today.

+ Single-core, the **performance** on these applications is increased **1.13x** with a **power** increase of **1.73x**. This compares with **Dual-core** where performance is increased **1.73x** with a power increase of **1.02x**.



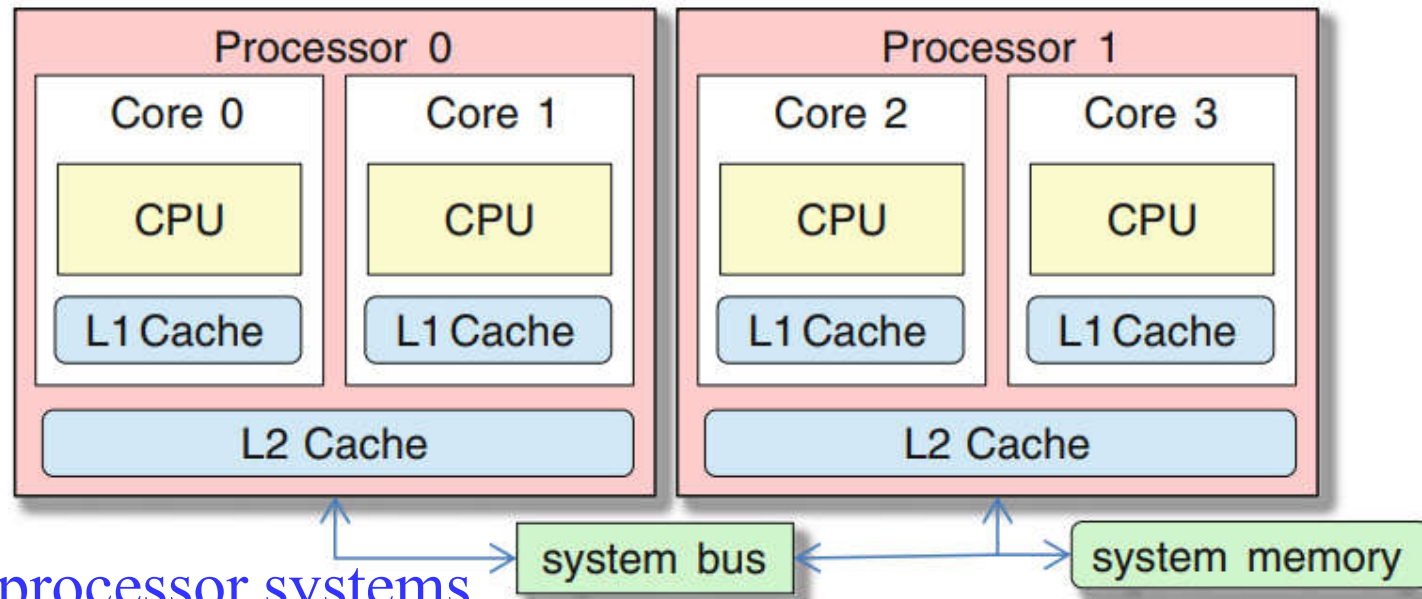
# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.2 Multi-core Processors

### 4.2.2 Multi-core Processor Architecture

**Multi-core processors** differ from **multiprocessor systems** physically as detailed in *Figure* and differ on a number of characteristics:

- + *Communication latency*
- + *Bandwidth*
- + *Number of processors*



→ It is *in contrast to* **Multi-processor systems**.

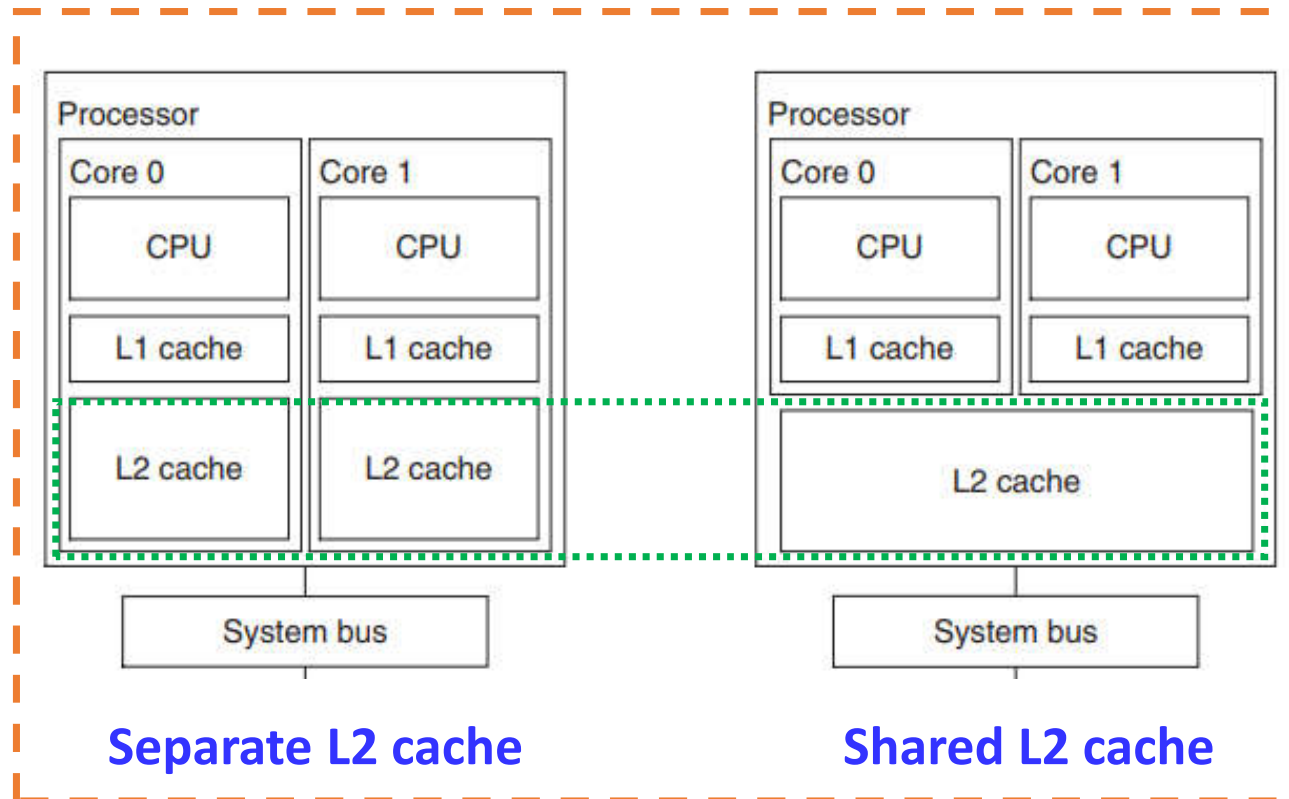
# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.2 Multi-core Processors

### 4.2.2 Multi-core Processor Architecture

The current trend is toward **shared level 2 cache** → Sharing of items between cores.

**Threads** on separate cores can synchronize through the **faster** level 2 cache **as opposed** to synchronization through main memory or the next level of cache.



# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



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## 4.2 Multi-core Processors

### 4.2.2 Multi-core Processor Architecture

#### □ Homogeneous and Heterogeneous Multi-core

- + A **homogeneous** multi-core processor is comprised of processor cores that support the **same** instruction set architecture (ISA).
- + A **heterogeneous** multi-core processor is comprised of processor cores that support **different ISAs**. → A *heterogeneous* multi-core processor has the advantage of employing ***specialized processor cores*** designed for the task.
- Advantages of *homogeneous* multi-core architectures: the design effort is **limited** and that software can **easily be migrated** from one processor to **another one**. This is very useful in case **one of the cores fails**.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.2 Multi-core Processors

### 4.2.2 Multi-core Processor Architecture

#### ❑ Symmetric and Asymmetric Multi-core

The *symmetric* and *asymmetric*, that differ in terms of:

- Memory access
- Operating system (OS)
- Communication between processor cores
- Load balance and processor affinity

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.2 Multi-core Processors

### 4.2.2 Multi-core Processor Architecture

#### ❑ Symmetric and Asymmetric Multi-core

##### ▪ Memory access

- + In **symmetric** multiprocessing (**SMP**), the processor cores in the system have similar views of the system and thus **share main memory**.
- + In **asymmetric** multiprocessing (**AMP**), the view of the system is different between processor cores. So memory regions could be disjoint, implying two processor cores **cannot directly access and modify** the same memory location.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



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## 4.2 Multi-core Processors

### 4.2.2 Multi-core Processor Architecture

#### □ Symmetric and Asymmetric Multi-core

##### ▪ Operating system

- + **SMP** will have one operating system (**OS**) that controls the processor cores.
- + An embedded system configured as an **AMP** system may contain multiple **OSes** that **effectively partition** the system.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.2 Multi-core Processors

### 4.2.1 Multi-core Processor Architecture

#### ❑ Symmetric and Asymmetric Multi-core

##### ▪ Communication between processor cores

+ **SMP**: Communication between processor cores is accomplished by **sharing memory** and relying upon the OS to provide functionality to **support synchronization and locking of memory**.

+ **AMP**: With disjoint memory regions, communication is accomplished through a **message passing facility** that serves to **marshal data** and **pass it from one OS to another** for use.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.2 Multi-core Processors

### 4.2.2 Multi-core Processor Architecture

#### ❑ Symmetric and Asymmetric Multi-core

##### ▪ Load balance and processor affinity

+ **SMP**: Load balancing is **accomplished by the OS** and subject to **its scheduling**. SMP-enabled OSes *support processor affinity*, which allows the assignment of processes to specific processor cores.

+ An **AMP** system is statically partitioned, so the processor cores assigned to one OS *will not be shared by a different OS* on the embedded system.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.2 Multi-core Processors

### 4.2.3 Benefits of Multi-core Processors

+ Multi-core processors offer developers the ability to apply more computer resources to a particular problem. These offer two types of advantages: improved **processing time** or increasing **throughput**.

+ By taking advantage of multi-core processors, any one server could handle a greater number of transactions in the **desired response time**. Balancing both improved **processing time** and increased **throughput** is power utilization.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



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## 4.3 Graphic Processing Units (GPUs)

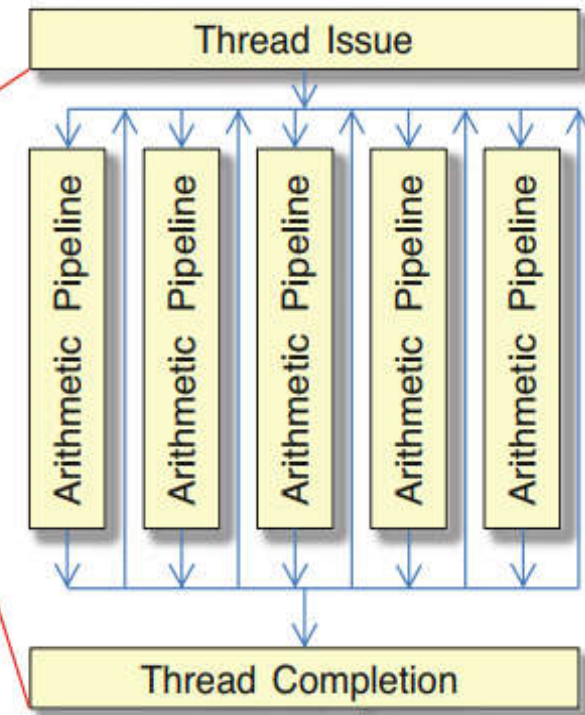
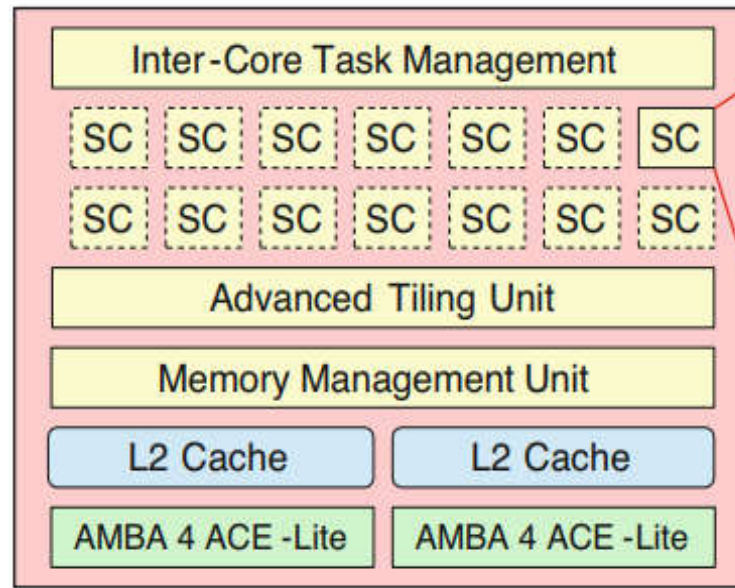
- + Many computers used specialized graphics processing units (**GPUs**) in order to generate an appealing graphical representation of computer output.
- + Current **GPUs** try to run a large number of computations concurrently in order to achieve the **desired performance**. The standard approach to concurrency is to **run many fine-grained threads at the same time**.
- The goal of **GPUs** is to *keep many processing units busy* and to **hide memory latencies** by **fast switching** between **threads**, in order to **achieves high performances** in an **energy-efficient way**.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.3 Graphic Processing Units (GPUs)

### + ARM Mali™ -T880 GPU

The number of **SC cores** is 16. **Each core** includes several pipelines for the execution of *arithmetic*, *load/store*, or *texture-related instructions*.

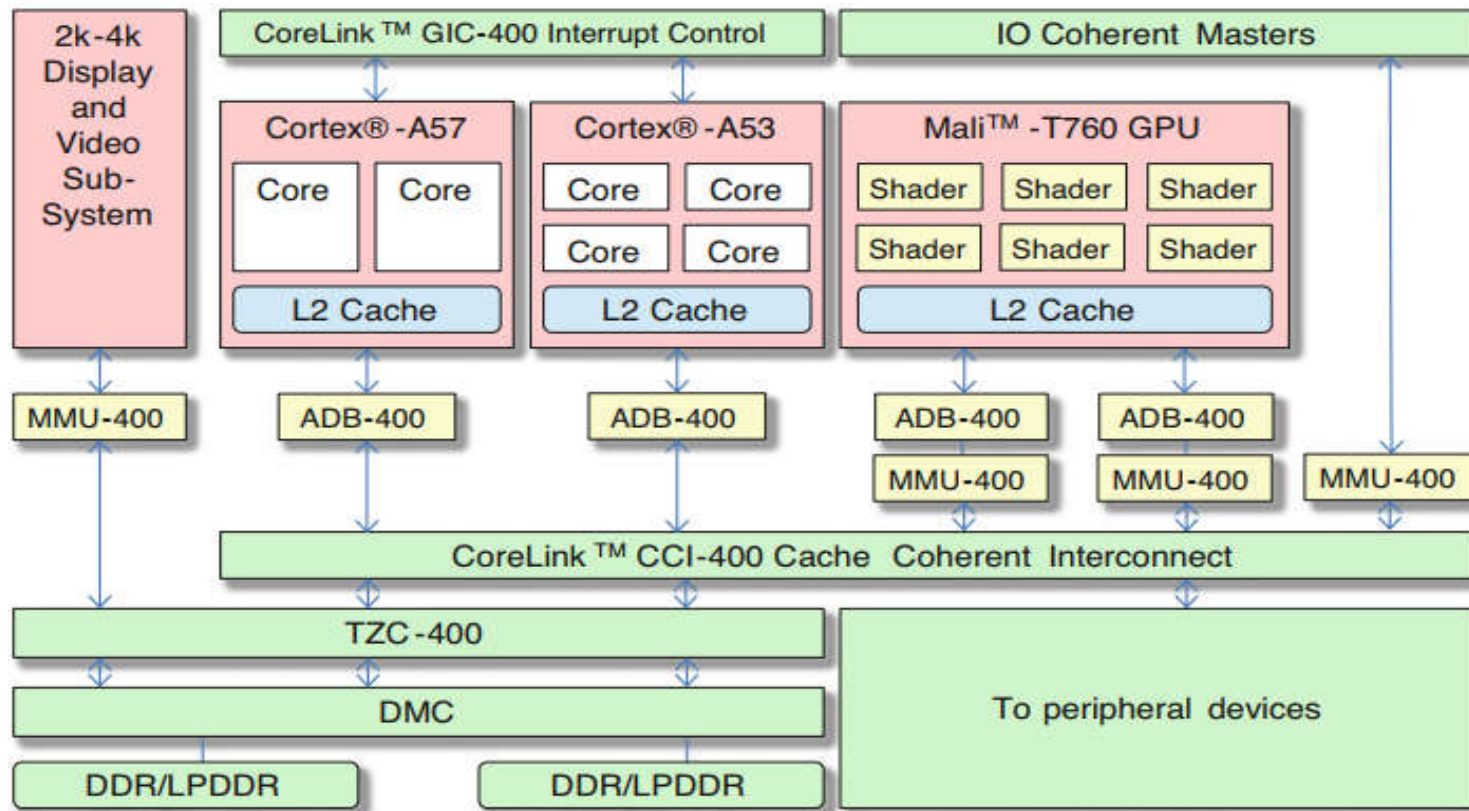


→ These **hardwired solution GPUs** are unable to support **non-standard computer graphics algorithms** and have been replaced by *programmable solutions*.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.4 Multiprocessor System on a Chip (MPSoC or SoC)

An **ARM heterogeneous multi-core system** have been merged with a GPU (**SoC**):



# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

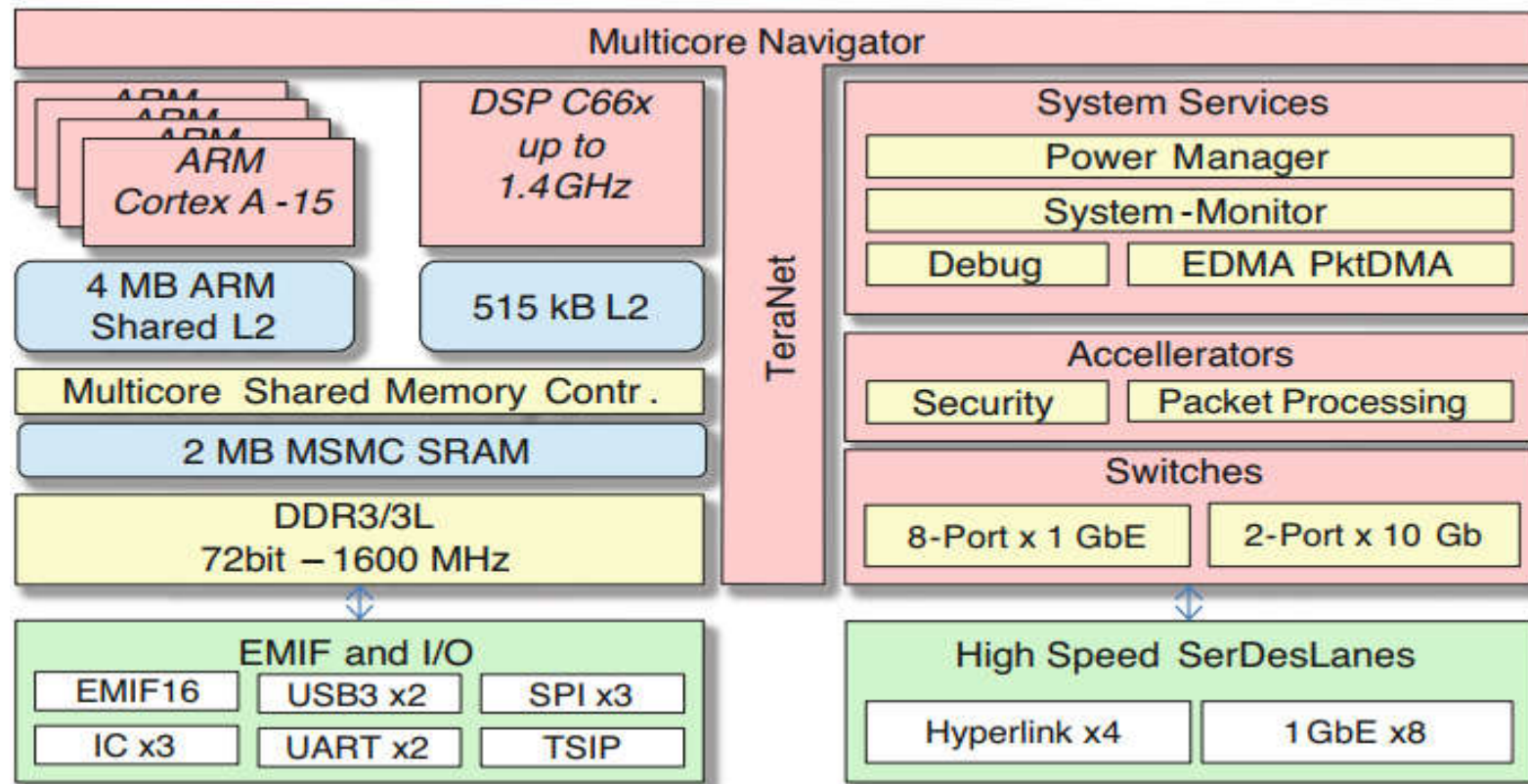
## 4.4 Multiprocessor System on a Chip (MPSoC or SoC)

- + The architecture shown in Figure does **not only contain processor cores**. it comprises a number of additional system components.
- + Overall, the idea behind this integration is **to avoid extra chips for such functionality**. As a result, a **whole system** is integrated on one chip.
- + Therefore, we are calling such an architecture a system-on-a-chip (**SoC**) or even a multiprocessor system-on-a-chip (**MPSoC**) architecture.
- The networks for **MPSoCs** will be fast and provide **lower-latency** communication between the processing elements, while **distributed systems (multiprocessors)** give **higher latencies than** are possible on a **single chip**.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.4 Multiprocessor System on a Chip (MPSoC or SoC)

An 66AK2x **MPSoCs** (*Texas Instruments*) contains **ARM and C66xx processors**



# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



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## 4.5 Field programmable gate arrays (FPGAs)

### ❑ Reconfigurable Logic

+ In many cases, full-custom hardware chips (ASICs) **are too expensive**, and **software based solutions are too slow** or **too energy-consuming**. Reconfigurable logic provides a solution if algorithms can be efficiently implemented in *custom hardware*.

+ It can be almost *as fast as Special-purpose hardware*, but in contrast to special purpose hardware, the performed function can be changed by using *configuration data*.

→ **FPGAs** are the *most common form of Reconfigurable hardware*, such devices are programmable “in the field”.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA

## 4.5 Field programmable gate arrays (FPGAs)

❑ **FPGAs.** The **Xilinx® UltraScale** architecture

+ Some columns contain *I/O interfaces*, *clock* devices, and/or *RAM*.

+ Other columns include **configurable logic blocks (CLBs)**, special hardware for *Digital Signal Processing (DSP)*, and some *RAM*.

+ **CLBs** are the key components, they provide **configurable functions**.

|                                       |
|---------------------------------------|
| Transceivers                          |
| CLB, DSP, Block RAM                   |
| I/O, Clocking, Memory Interface Logic |
| CLB, DSP, Block RAM                   |
| I/O, Clocking, Memory Interface Logic |
| CLB, DSP, Block RAM                   |
| Transceivers                          |

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



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## 4.5 Field programmable gate arrays (FPGAs)

### ❑ The Xilinx® UltraScale architecture:

- + Each **CLB** contains eight blocks. Each **block** includes a RAM which is used to implement logic functions.
- + **Blocks in a CLB** can be combined to form adders, multiplexers, shift registers, or memories.
- + **Configuration data** determines **the setting of** multiplexers in **the CLBs**, the clocking of registers and RAM, the content of RAM components, and the connections between CLBs.
- + Several **CLBs** **can be combined** to create adders having a larger bit width, memories having a larger capacity, or complex logic function.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



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## 4.5 Field programmable gate arrays (FPGAs)

### □ Designs of FPGA

- + Currently available FPGAs comprise a large number of **specialized blocks**, like hardware for *digital signal processing* (DSP), *some memory*, *high-speed I/O devices* for various I/O standards, *a decryption facility* for FPGA *configuration data*, *debugging support*, *ADCs*, *high-speed clocking*, etc.
- + Integration of reconfigurable computing with processors and software is simplified **if processors are available** in the FPGAs. There may be either **hard cores** or **soft cores**.

# 4. CÔNG NGHỆ Multi-Core, SoC, FPGA



CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

## 4.5 Field programmable gate arrays (FPGAs)

### ❑ Designs of FPGA

- For **hard cores**, the layout contains a **special area** implementing a core in a dense way. This area **cannot be used for anything** but the hard core.
  - **Soft cores** are available as synthesizable models which are mapped to standard CLBs. Soft cores **are more flexible** but **less efficient** than **hard cores**. Soft cores can be implemented on any FPGA chip.
- Overall, FPGAs allow implementing a **huge variety** of **hardware devices** **without any need to create hardware** other than FPGA boards.

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